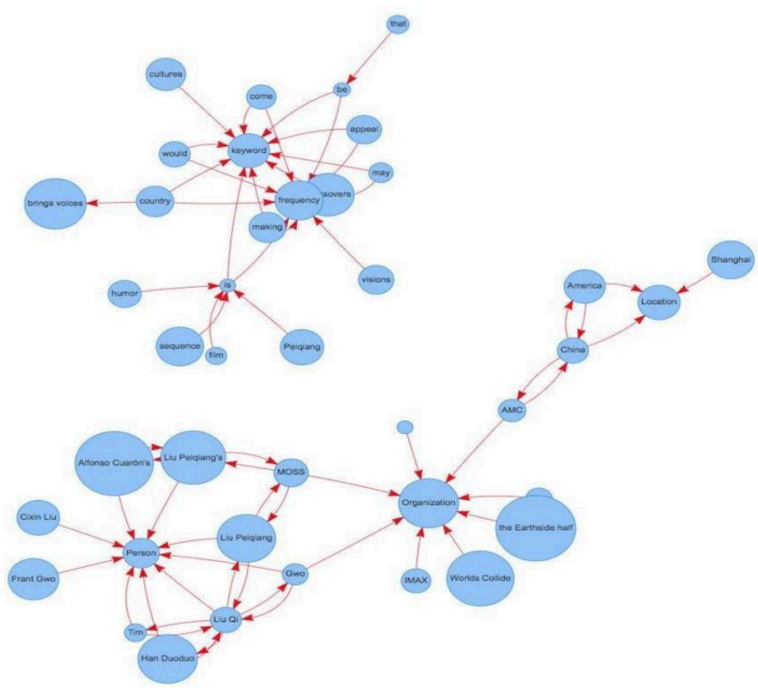
	Marwadi University **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** Representation of a document with their keywords using TextRank	
Experiment No: 09	**Date:**	**Enrolment No:92301733038**

<u>Aim:</u> Representation of a document with their keywords using TextRank

<u>IDE:</u> Google Colab

<u>Theory:</u>

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.



Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank.

PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

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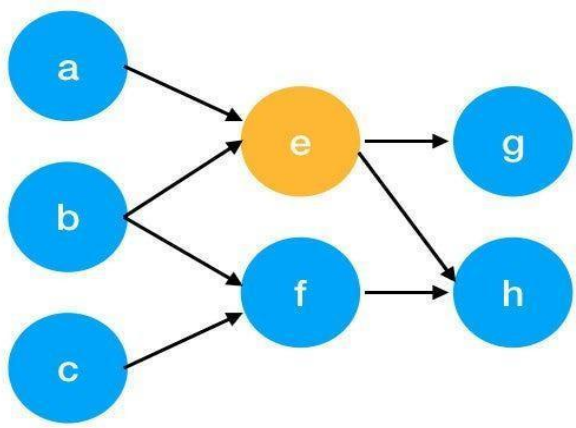
$$S(V_i) = (1-d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- S(Vi) - the weight of webpage i
- d - damping factor, in case of no outgoing links
- In(Vi) - inbound links of i, which is a set
- Out(Vj) - outgoing links of j, which is a set
- |Out(Vj)| - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above

$$In(v_e) = \{a, b\}\, , j \in \{a, b\}$$

$$\sum_{j \in \{a,b\}} \frac{1}{|Out(V_j)|} S(V_j) = \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b)$$

$$= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b)$$

$$= S(V_a) + \frac{1}{2} S(V_b)$$

function to a simpler version.

We can get the weight of webpage e by the following function.

$$S(V_e) = (1-d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

Pre Lab Exercise:

1. What are the key components of TextRank?
- Text preprocessing (tokenization, stopwords removal)
- Graph construction (words as nodes, co-occurrences as edges)
- PageRank algorithm application
- Ranking and extraction of top keywords

2. What are some limitations of TextRank for keyword extraction?
- No semantic understanding
- Sensitive to window size
- Struggles with multi-word phrases
- Ignores part-of-speech (POS) tags

3. What are the advantages of TextRank for keyword extraction?
- Unsupervised (no training data needed)
- Language-independent
- Simple and efficient
- Considers global word importance
- Robust and adaptable

Program (Code):

```
import nltk
import networkx as nx
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import string

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('punkt_tab')

def textrank_keywords(text, window_size=4, top_n=5):

    words = word_tokenize(text.lower())

    stop_words = set(stopwords.words('english'))
    words = [w for w in words if w not in stop_words and w not in
string.punctuation]

    graph = nx.Graph()

    for i, word in enumerate(words):
        for j in range(i+1, i+window_size):
            if j < len(words):
                graph.add_edge(word, words[j])

    scores = nx.pagerank(graph)

    ranked_words = sorted(scores.items(), key=lambda x: x[1],
reverse=True)

    keywords = [word for word, score in ranked_words[:top_n]]

    return keywords


text = """
```

```
Machine learning is a field of artificial intelligence that uses
statistical techniques
to give computer systems the ability to learn from data without being
explicitly programmed.
"""
```

```
keywords = textrank_keywords(text)
```

```
print("Extracted Keywords:", keywords)
```

Results:

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
Extracted Keywords: ['artificial', 'data', 'intelligence', 'learn', 'uses']
```

Observation:

TextRank identifies important words by analyzing their connections and co-occurrences within a text, similar to how PageRank ranks web pages

Post Lab Exercise:

1. Write about "your ownself" (in not more than 500 words ☾ You know better about you, rather than ChatGPT!!). Extract top-25 keywords for your portfolio and analyze it in 3 categories: keyword can not give the real idea, keyword gives the real idea, keyword is irrelevant. Paste the code, your portfolio, output and your analysis.

CODE:-

```
my_portfolio_text = """
<!-- Paste your 'about yourself' text here. For example: -->
Hello, I am Meet Limbani, Currently i am doing Btech in ICT Department in
Marwadi university . I enjoy Play Cricate and building practical solutions
that help solve real-world problems.
```

```
I have developed projects such as an AI-based Plant Disease Detection
System and a Farmer Support Dashboard, which aim to assist farmers in
identifying crop diseases and improving productivity."""

# Extract top 25 keywords
portfolio_keywords = textrank_keywords(my_portfolio_text, top_n=25)

print("Top 25 Keywords from your portfolio:")
print(portfolio_keywords)
```

OUTPUT:-

```
Top 25 Keywords :
['crop', 'identifying', 'example', 'farmers', 'assist', 'hello', 'aim', 'meet',
'dashboard', 'limbani', 'support', 'currently', 'farmer', 'btech', 'system',
'ict', 'detection', 'department', 'disease', 'marwadi', 'plant', 'university',
'ai-based', 'enjoy']
```